AI-101 MINI PROJECT · GROUP 84

DateWise

MATCH THEORY — The AI Dating Strategist

Predicting mutual preferences from speed-dating data with a transparent machine-learning baseline.

PROJECT REPOSITORY

github.com/Srija3110/AI-Project_84

SCHOOL OF AI & DATA ENGINEERING · IIT ROPAR

THE TEAM

Ishant Sharma

Execution Lead · 2026AIB1043

Sarika Naga Sai Srija

Documentation Lead · 2026EEB1407

Jatin

Video Lead · 2026EEB1358

Kavita Meena

PPT Lead · 2026AIB1049

The problem and the data

WHAT WE PREDICT

Can pre-date information help estimate whether two participants will mutually choose each other? A match is two-sided: both participants must indicate interest.

WHY IT MATTERS

We combine information from both participants symmetrically to estimate whether their choices form a mutual match.

8,378
directed records

195
raw columns

4,184
reciprocal pairs

21
speed-dating waves

DATASET & TARGET

Kaggle Speed Dating Experiment · Speed Dating Data.csv
0 = no mutual match | 1 = mutual match
Pair construction: reverse-direction records paired; one row per unordered pair retained.
Class imbalance: 690 / 4,184 pairs (≈16.5%) were mutual matches.

CLASS DISTRIBUTION · 3,494 non-matches | 690 mutual matches

Our approach

1

Load data

Read speed-dating records

→

2

Pair records

Create reciprocal unordered pairs

→

3

Build features

Means, absolute differences & agreement

→

4

Preprocess

Median imputation + standardisation

→

5

Train & evaluate

Balanced Logistic Regression

LIBRARIES USED

Python · Pandas · NumPy · Scikit-learn
Matplotlib · Seaborn

MODEL & VALIDATION

80:20 GroupShuffleSplit by wave
Balanced Logistic Regression · max_iter = 1,000
Decision threshold = 0.40

IMPLEMENTATION

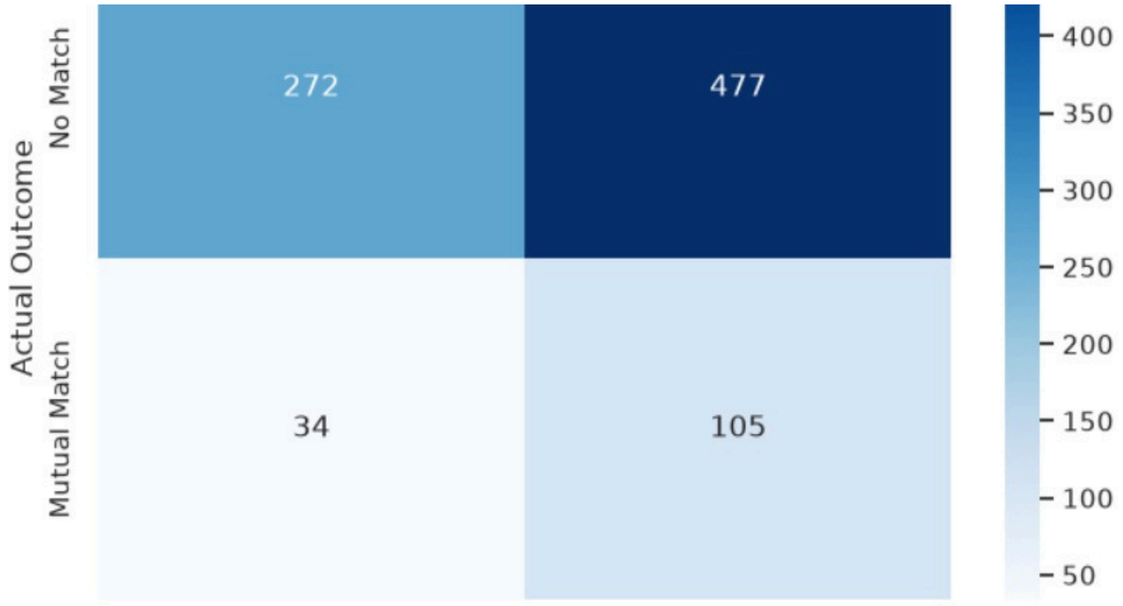
main.py runs the workflow and creates the output folder. Stable matching is future scope—not part of the current executable baseline.

Results

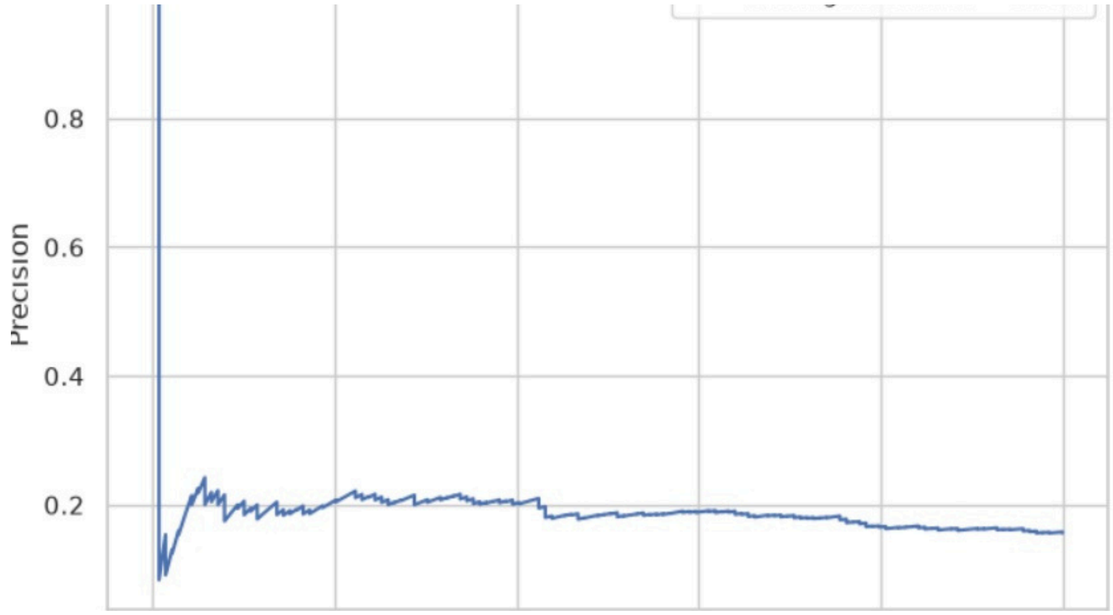
FIRST MODEL (BEFORE BALANCING) **82.9%** accuracy · ≈1% recall

FINAL MODEL (CLASS-BALANCED, THRESHOLD 0.40) · TEST SET

42.45%	18.04%	75.54%	29.13%	0.5701	0.1937
Accuracy	Precision	Recall	F1-score	ROC–AUC	Avg. Precision



CONFUSION MATRIX



PRECISION–RECALL CURVE

TN = 272 · FP = 477 · FN = 34 · TP = 105

Recall is high, but many predicted matches are false positives. ROC–AUC 0.5701 indicates modest ranking ability. The first model reached 82.9% accuracy but only ≈1% recall—it mostly predicted “no match”; class balancing improved minority-class detection at the cost of accuracy.

Learnings & team responsibilities

WHAT WORKED

- Class-weight balancing improved minority-class detection.
- Threshold tuning (0.40) raised recall to 75.54%.
- Pair features represent both participants symmetrically.

WHAT DIDN'T

- Precision is 18.04%: many false positives remain.
- Accuracy is 42.45%; ROC–AUC is 0.5701.
- Positive-match identification remains difficult (AP 0.1937).
- Accuracy can be misleading: the first model scored 82.9% but caught only ≈1% of matches, since ~83.5% of pairs are non-matches.

WHO DID WHAT

Sarika Naga Sai Srija
Documentation · GitHub repo work · Final plots · Model training

Ishant Sharma
Execution Lead · Dataset · Exploration · Code · Feature engineering

Kavita Meena
PPT Lead · Model evaluation

Jatin
Video Lead · Data cleaning

NEXT STEPS

Improve pair features, compare additional models, validate across more splits and explore preference-based / stable matching as future scope.

AI-USE DISCLOSURE · AI assisted feature-engineering exploration, metric interpretation, workflow/debugging explanations and presentation wording. The team reviewed the project claims and results. Predictions reflect one historical experiment—not universal compatibility or a judgement about people.